

C02. Topics in Nonlinear Dynamics

2.1 The Cardiovascular System

Applications of nonlinear dynamics to the cardiovascular system are typically related to the electrophysiology of the heart such that the phenomenon of interest is usually events that are of the time intervals less than 1 min. Of particular interest is the heart rate variation and arrhythmias. Nonlinear methods provide a different perspective into electrophysiology and are typically not taught in the classical medical physiology textbook. In particular, physiology can explain and identify the origin of some arrhythmias, but the most common, the sinus arrhythmia, remains to be explained. The cardiac rhythm was generally introduced in Chap. 1 as the concept of the bio-oscillator. A nonlinear oscillator may be used to analyze many known cardiac rhythms normal and pathological. Moreover, since the cardiac pacemaker is a repetitive process, the concept of the Fractal can be employed. Cardiac electrophysiology has enjoyed a long history of research, so that information is available at the heart level, the cellular level, and the molecular level. Nonlinear dynamics will be employed in this book to revisit each structural level with a new vantage point. A focal point of debate is the origin of cardiac fibrillation. Briefly, fibrillation is known as an uncoordinated electrical and mechanical activation of the cardiac muscle that leads to death due to net zero pumping of blood. Some dynamicists have theorized that fibrillation is Chaos. These discussions will be covered in more detail in later chapters after the reader becomes more familiar with nonlinear dynamics. Continuing along with our survey of nonlinear dynamics in physiology, move next to the pulmonary system.

2.2 Respiratory Physiology

Recall from Chap. 1 that the Fractal concept applies well to the modeling of the geometry of the respiratory system. Also, the control of breathing lends itself well to the application of nonlinear dynamic methods. Of course, everyone is familiar with breathing being a periodic process. It would be simple to suggest that breathing is stimulated via pacemaker cells just as in the heart. Surprisingly, the respiratory system possesses no such respiratory oscillator. Instead, the use of a nonlinear feedback control system explains the source of oscillations (Glass & Mackey, 1988). Physiologists are familiar with

the homeostatic reflex regulation of breathing. Reflex regulation is a negative feedback control system that is employed extensively in human physiology. By adding nonlinear affecters and feedback delay, it is possible to cause the respiratory regulation system to become unstable and oscillate. This kind of treatment of the respiratory system explains the source of oscillatory breathing. This approach further explains some pathology of breathing such as respiratory arrest and Cheyne's stokes phenomenon and other abnormal breathing patterns.

2.3 Population Dynamics

Probably one of the first applications of nonlinear dynamics to biology is that of the study of population dynamics, that is, to analyze how various animal populations vary in time. Of recent interest is the study of how diseases may spread through a population. The analysis of disease spread is a subject of epidemiology population dynamics analysis applied well in the case of a confined population. For example, a human population that is constrained within the borders of a country. Many readers are familiar with epidemiologist presentations of the spread of COVID infections during 2020 (Giordano et al., 2020). The theory behind the infection-time curves rests on the analysis of population dynamics. Additionally, population dynamics may also be applied to populations of cells. For example, the dynamics of the human blood cell population. Human cell dynamics is useful to help analyze various types of anemia in white and red cell concentration. The analytical knowledge of these cell populations can be used for more effective treatment of disease that causes extreme low-cell concentrations.

In summary, nonlinear dynamics analysis applies well to much of human physiology. We will explore more of these applications as we move through this book. At this time, it is useful to examine some of the analytical methods that are valuable in our study such as basic population models.

2.4 Population Model Solution

To illustrate some of the fundamental methods by which the nonlinear biological system may be analyzed, a simple population model is now introduced and solved. A characteristic feature of unrestrained population growth is that the rate of growth is proportionate to the magnitude of the population

itself. Mathematically, this can be written as

$$P'(t) = aP(t)$$

where $P(t)$ is the population versus time, and P' is its time derivative; “ a ” is a constant of proportionality that must be calibrated to the population. The solution of this differential equation is easily determined to be

$$P = PP_0e^{at}$$

The constant P_0 is the initial population. Following this model, it is found that the population is unstable, that is, as t increases, the population increases exponentially to infinity for $a > 0$. Reexamining the model for the case where $a < 0$, it is found that the model would predict that population decreases exponentially to zero as t increases. Hence, for either sign of the model, results are unreasonable. A sample population growth curve is shown in Fig. 2.1 for a positive value of a , where infinite population growth occurs.

This model is flawed in the final value prediction of the model. There are only two final values: infinite population or a zero population. The model is flawed in these outcomes, but the model predictions are not unreasonable over short time intervals. So, rather than abandoning the model completely, look at a modification. Returning to Eq. 2.1 of the model, it is a linear model that permits the population to grow in direct proportion to its size. This problem can be repaired by adding a term that prevents growth when the population becomes too large. So, modify the population model now to limit growth according to a new differential equation,

$$\frac{dP}{dt} = aP(L - P)$$

where L is the population limit. Here, we will use a normal value of $L = 1$. For Eq. 2.3, it can be seen that as the population grows larger than L , the derivative of population growth will turn negative and prevent further growth. Or, when $L = P$, $dP/dt = 0$, so that the steady state P of this new model is for population to converge to L . Let's confirm this by solving Eq. 2.3 for $P(t)$. At this point, we run into a new problem. That is, after expanding the right side of Eq. 2.3 we see that a term aP^2 emerges to result in a nonlinear differential equation. Hence, the reader now sees our first

example of a nonlinear dynamic system in this book. There is much to learn from this very simple nonlinear model. To solve this equation, Euler method is now introduced. This will be a computational solution of the differential equation.

Start by defining two adjacent population points in time: population at the current time is $P(t_i)$, then, after a small change in time Δt , $P_{i+1} = P(t_i + \Delta t)$. Assuming that the time change can be made infinitely small, the derivative of population then can be written as

$$\frac{dP}{dt} = \frac{P_{i+1} - P_i}{\Delta t}$$

Using the definitions and substituting them into Eq. 2.3, the new model is converted to Euler computational form to be Eq. 2.4:

$$P_{i+1} = \Delta t a P_i (1 - P_i) + P_i$$

This form of our new population model is known to those who study nonlinear dynamics or population modeling as the logistic equation [REF]. Equation 2.4 is now in the form such that we enter the current population value P_i and simply compute the next value of population P_{i+1} . Hence, we now have an iterative solution for the original differential equation model Eq. 2.4. A sample iteration of Eq. 2.4 is provided in Fig. 2.2. The properties of this equation are very illustrative of nonlinear dynamic systems. These features will be examined in more detail in a later chapter, but for now notice that the new population model is functioning as required. In Fig. 2.2, the vertical axis represents population, and the horizontal axis is time, where each value n indicates a time step of Δt . Observe that population now converges to a specific level of about 0.5. As you 'll find later, no matter what the initial population starts at, this model will converge to the .5 level. This was determined by the choice of parameter $a = 2$.

The vertical axes represent population value and the horizontal axes represent time. In this graph, a value of $a = 2$ and initial population of 0.6 were used.